

Sequence Modeling: RNNs and the Temporal Dimension

Scaler School of Technology | Deep Learning I | Week 5 — Session 2/2 | May 2026



What Is This Lecture About?

Last session you learned transfer learning and practical CNN applications. This session we **leave images behind and enter the world of sequences** — text, audio, time series. You'll learn how Recurrent Neural Networks (RNNs) process data where *order matters*, and discover both their power and their fundamental limitation.

1 The Vanilla RNN

A neural network with a loop — it reads one element at a time and maintains a “hidden state” that acts as memory. *Think of it like reading a book one word at a time while keeping a mental summary that updates with each page.*

2 Backpropagation Through Time (BPTT)

How RNNs learn: unroll the loop across all timesteps, then apply standard backprop on the resulting deep graph. *Like unrolling a spiral staircase into a long hallway so you can see the entire path at once.*

3 Vanishing Gradients and RNN Configurations

Why RNNs struggle with long sequences — gradients shrink exponentially over time. Plus: the different ways to wire an RNN (many-to-one, many-to-many) for tasks like sentiment analysis and tagging.

What to Expect in the Session

- Build a vanilla RNN from scratch in numpy
- Visualize how hidden states evolve over time
- See vanishing gradients in action (real gradient magnitudes!)
- ★ Train a character-level RNN that predicts the next letter
- Short quizzes after each major section
- Compare many-to-one vs. many-to-many architectures

How to Prepare (Optional but Helpful)

- ✓ **Recall** the chain rule from backprop (Week 2)
- ✓ **Think about:** what makes a sentence different from an image as input to a neural network?
- ✓ **Skim** matrix multiplication refresher — we'll see repeated matrix products
- ✓ **Optional:** try typing `nn.RNN` in PyTorch docs to see the interface

Duration: ~2 hours **Prerequisites:** Backpropagation, computational graphs, basic PyTorch **Instructor:** Priyansh Saxena